

Digital Resource Creation for Bishnupriya Manipuri: Corpus Development and Computational Perspectives for a Low-Resource Language

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Abstract

The development of language resources plays a crucial role in advancing computational linguistics, natural language processing (NLP), and digital language preservation. However, many indigenous and endangered languages continue to remain underrepresented in the digital ecosystem due to the absence of structured linguistic datasets. Bishnupriya Manipuri, an Indo-Aryan language spoken primarily in parts of Northeast India and Bangladesh, is one such low resource language facing significant challenges in digital representation and computational processing. This study presents the design and development of a standardized digital corpus for Bishnupriya Manipuri by integrating textual data collected from literary works, cultural documents, educational materials, folk narratives, and online resources. Particular attention has been given to the conversion of legacy ASCII-based texts into Unicode-compliant formats to ensure interoperability with modern computational tools. The corpus development process involved data acquisition, text normalization, encoding standardization, sentence segmentation, tokenization, and structured storage. The resulting corpus comprises approximately 100,000 words and 10,196 sentences, representing diverse genres and linguistic styles. Corpus based analysis has been performed to examine lexical frequency patterns, morphological variations, and grammatical structures. The developed corpus establishes a foundation for future Natural Language Processing (NLP) applications, including part-of-speech tagging, morphological analysis, language modelling, and other computational language technologies for the Bishnupriya Manipuri Language. It establishes a foundation for applications such as corpus annotation, morphological studies, language modelling, and other computational tools for low-resource languages. The work contributes to ongoing efforts in language preservation by providing a scalable linguistic resource for Bishnupriya Manipuri and establishing a framework that can be adapted for other under resourced languages. The developed corpus is expected to support future interdisciplinary research in corpus linguistics, computational linguistics, digital humanities, and indigenous language technology development.

Keywords: Bishnupriya Manipuri, Corpus Linguistics, Digital Resource Creation, Low-Resource Language, Language Preservation, Computational Linguistics

I. Introduction

A corpus is a systematically organized and electronically stored collection of texts used for linguistic research and computational language processing. Corpora play a crucial role in developing language technologies such as spell checkers, search engines, speech recognition systems, machine translation tools, and educational applications. For widely spoken languages, large-scale annotated corpora are readily available. However, for many indigenous and endangered languages, including Bishnupriya Manipuri, such resources are scarce or nonexistent.

Bishnupriya Manipuri has traditionally been studied from historical, sociolinguistic, and morphological perspectives. Computational linguistic exploration of the language remains minimal due to the lack of digitized textual data. The pioneering work by [5] marked the first effort toward corpus development for Bishnupriya Manipuri. The present paper extends that work by enriching the corpus, refining

methodological approaches, and integrating new analytical dimensions relevant to contemporary NLP research.

Objectives of the Study

- i. To create a standardized digital language resource for Bishnupriya Manipuri through the collection, digitization, and organization of textual data from diverse sources.
- ii. To analyze the linguistic and morphological characteristics of Bishnupriya Manipuri through corpus-based methods, including tokenization, frequency analysis, and examination of grammatical structures.
- iii. To create a computationally usable digital language resource that establishes a foundation for future Natural Language Processing (NLP) applications, including part-of-speech tagging, morphological analysis, language modelling, and other language technology tools for the Bishnupriya Manipuri language.

This paper is structured as follows: Section II provides a linguistic overview of Bishnupriya Manipuri. Section III discusses literature and research design. Section IV presents linguistic and morphological analysis. Section V focuses on frequency analysis and quantitative observations. Section VI highlights major contributions of the study. Section VII concludes the paper and outlines future research directions.

II. Overview of the Bishnupriya Manipuri Language

Bishnupriya Manipuri is an Indo-Aryan language historically spoken around the Loktak Lake region of Manipur. The language is closely related to Assamese, Bengali, and Oriya. This linguistic hybridity makes Bishnupriya Manipuri particularly interesting from both typological and computational perspectives.

Despite its historical roots, Bishnupriya Manipuri is now nearly extinct in its place of origin and is primarily preserved by diaspora communities. [12] has classified it as an endangered language, underscoring the urgency of documentation and digital preservation.

While significant descriptive linguistic work has been carried out by scholars such as [3,4], especially in morphology and etymology, computational resources for the language remain extremely limited. This research seeks to address that gap.

III. Literature

For any deep computational processing of language evidences had been searched, and one such set of evidences is corpus. This paper describes the development of a text based corpus for the Bishnupriya Manipuri language. A Corpus is considered as a building block for any language processing tasks. Due to the lack of awareness like other Indian languages, it had been also studied less frequently. As a result the language still lacks a good corpus and basic language processing tools. As per knowledge this is the first effort to develop a corpus for Bishnupriya Manipuri language[6].

In this work a morphological analysis of Bishnupriya Manipuri language has been presented, an Indo-Aryan language spoken in the north eastern India. As of now, there is no computational work available for the language. Finite state morphology is one of the successful approaches applied in a wide variety of languages over the year. Therefore the finite state approach had been adapted to analyse morphology of the Bishnupriya Manipuri language[5]. Similar finite state transducer methods have proven effective for other low-resource Indo-Aryan languages like Maithili, highlighting their suitability for morphologically rich languages with limited data[9].

This author conveys a case of systematic survey of outstanding progress on corpora conducted by researchers affiliated with different common section institutions all over the world. Such a range overview selected 20 outstanding types of research from multi research pushing institutions all around the world. These projects employ corpus techniques and technology to treat an enormous domain of research queries that are relevant to linguistic studies, language teaching and learning, cultural studies, and discourse analysis. These varied implementations of corpus techniques and advances clearly explain the great stress and chances that corpora applied in linguistics can hand to those who have the intention to research, educate, and learn the language[1].

Recent studies indicate that NLP research has predominantly focused on high-resource languages, while low-resource languages continue to face challenges due to limited annotated corpora, linguistic resources, and computational tools. To address these limitations, researchers have explored transfer learning, multilingual transformer models (e.g., mBERT and XLM-R), and cross-lingual learning for tasks such as part-of-speech tagging, morphological analysis, machine translation, and language identification. Developing digital corpora remains essential for enabling NLP applications, preserving linguistic diversity, and promoting digital inclusion for underrepresented languages[8].

A recent survey of Large Multimodal Models (LMMs) for low-resource languages reviewed 117 studies covering 96 languages and found that visual information, cross-modal transfer, and multilingual learning significantly improve NLP performance in data-scarce settings. The study also highlighted persistent challenges, including limited training data, computational efficiency, and hallucination, emphasizing the need for richer language resources and efficient multimodal approaches to support underrepresented languages[7].

Recent studies emphasize that NLP advancements have largely benefited high-resource languages, leaving many low-resource languages with limited access to modern language technologies. Researchers advocate culturally informed and community-driven NLP approaches that combine transfer learning, multilingual language models, active learning, and few-shot learning while respecting linguistic diversity and local cultural contexts. The study highlights that sustainable language resource development and community participation are essential for creating inclusive and effective NLP solutions for underrepresented languages[11].

The corpus comprises total words of approximately 100,000, total sentences of 10,196, and is monolingual and text-based in the Bengali-Assamese script (Unicode). Efforts have been made to ensure representativeness across genres, balanced inclusion of narrative, descriptive, and informational texts and sentence-level segmentation for future annotation. These challenges in data sourcing and normalization are common in low resource Indic languages, where script variation and data scarcity persist[10].

Although previous studies have initiated corpus development and morphological analysis for Bishnupriya Manipuri, existing resources remain limited in corpus size, Unicode conversion absent, no tokenization, no NLP ready structure and no statistical analysis. The present study seeks to bridge these gaps through the development of an expanded, standardized, and computationally usable corpus.

III.1 Research Design

Due to limited online presence, corpus development for Bishnupriya Manipuri required sourcing data from multiple channels like [2], digitized literary texts and folk narratives, educational materials and essays and legacy documents written in Smriti (ASCII-based) font. A major challenge has been the

prevalence of non Unicode ASCII fonts. To address this, an ASCII-to-Unicode (UTF-8) conversion tool has been developed and used to convert approximately 45,000 words. This step ensured interoperability with modern NLP tools and platforms.

The multi-source collection strategy improved linguistic diversity and enhanced the representativeness of the corpus.

Example of Raw Data (Before Processing)

- ASCII Text (Smriti Font):
amar logot jao
- Unicode Converted Text:
আমার লগত যাও

This conversion step has been essential to make the data compatible with modern NLP tools.

Normalization and Cleaning : The collected textual data have been cleaned by removing encoding inconsistencies, correcting spelling variations, eliminating duplicate entries, and standardizing orthographic forms.

Each document has been segmented into individual sentences before tokenization to facilitate downstream linguistic analysis.

Example:

- Sentence:
আমার লগত যাও। তুমি কি করিয়া আছ?
- Tokenized Output:
[আমার, লগত, যাও, ।, তুমি, কি, করিয়া, আছ, ?]

Table1: Corpus Statistics Table

Parameter	Value
Total Words	100,000
Total Sentences	10,196
Script	Bengali-Assamese Unicode
Corpus Type	Monolingual
Converted Legacy Words	45,000

This structured pipeline ensures that the corpus is machine-readable, scalable, and reusable for downstream NLP applications.

III.2 Corpus Processing Pipeline (with Flowchart)

The development of a high-quality linguistic corpus requires a systematic and multi-stage workflow that ensures the accuracy, consistency, and computational usability of the collected data. Figure 1 illustrates the corpus development and NLP processing pipeline adopted in the present study. The pipeline consists of several interconnected stages beginning with data acquisition and ending with the preparation of resources suitable for advanced Natural Language Processing (NLP) applications.

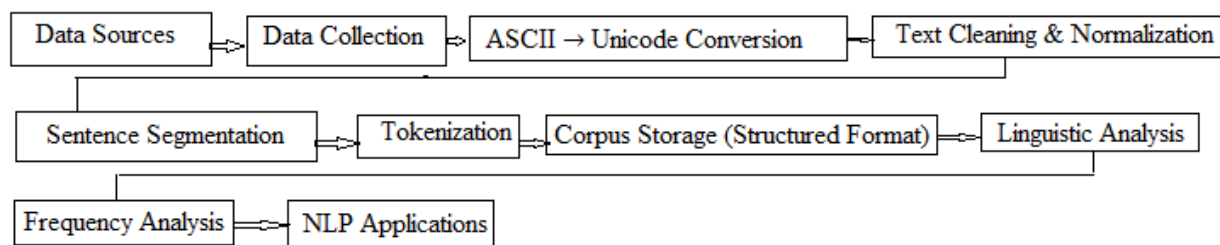


Figure 1. Corpus Development and NLP Processing Pipeline for Bishnupriya Manipuri Language Resource Development.

Figure 1 illustrates the corpus development pipeline adopted for Bishnupriya Manipuri. Textual data have been collected from diverse sources, including Wikipedia, literary works, folk narratives, educational materials, cultural publications, and archival documents. Legacy ASCII-encoded texts have been converted to Unicode, followed by text cleaning, normalization, sentence segmentation, and tokenization to produce a structured, machine-readable corpus. The processed corpus is then stored in a standardized format for linguistic and frequency analysis, providing empirical insights into the language's lexical and morphological characteristics. Finally, the corpus has been prepared for NLP applications such as part-of-speech tagging, morphological analysis, language modelling, spell checking, and machine translation.

IV. Linguistic and Morphological Analysis

Using the compiled corpus, detailed word-level analysis has been conducted, confirming and extending earlier grammatical descriptions. Observations have been made on noun inflections (gender, number, case), pronoun paradigms and verb morphology across tense, mood, and aspect. The corpus based evidence validates traditional grammatical rules while also revealing usage frequency and contextual variations not easily observable in manual studies. The corpus enabled a detailed empirical study of Bishnupriya Manipuri morphology, revealing both traditional grammatical structures and real-world usage patterns.

1. Noun Morphology : Nouns exhibit inflection based on number and case.

Example:

- Root: মানুহ (manuh – person)
- Plural: মানুহৰা (manuh-ra – people)
- Genitive: মানুহৰ (manuh-or – of the person)

2. Verb Morphology: Verbs display rich inflection across tense and aspect.

Example:

- Root: কৰ (do)
- Present: কৰে (does)
- Past participle: কৰিয়া (having done)
- Progressive: কৰিতেছে (is doing)

Observation: The corpus reveals that non-finite verb forms (e.g., কৰিয়া) occur frequently in compound sentence constructions, indicating a preference for complex clause structures in natural usage.

V. Frequency Analysis

Frequency analysis has been conducted to examine lexical distribution and identify commonly occurring functional words in the corpus. Approximately 25,000 inflected word forms have been identified, reflecting the rich morphological structure of Bishnupriya Manipuri. High-frequency items have been predominantly pronouns, conjunctions, auxiliary verbs, and postpositions, indicating their central role in sentence construction and discourse organization. The resulting frequency information provides a useful foundation for lexicon development, stop-word identification, part-of-speech tagging, and other NLP applications.

• **Table 2. Sample Frequency Distribution**

Word	Category	Frequency	Function
আমার	Pronoun	High	Possession marker
বারো	Conjunction	High	Clause linker
কৰিয়া	Verb	Medium	Non-finite verb
লগত	Postposition	Medium	Relational marker

The predominance of functional words and the large number of inflected forms demonstrate the suitability of the corpus for computational linguistic analysis and future language technology development.

Interpretation : High-frequency functional words dominate the corpus, reflecting natural grammatical structure rather than lexical richness. The presence of numerous inflected forms (~25,000) indicates morphological productivity. These patterns are critical for Stop-word identification, Language modelling, POS tagging

VI. Major Contributions of the Study

The present study extends previous work on Bishnupriya Manipuri corpus development by providing an expanded, standardized, and computationally usable linguistic resource. The major contributions are summarized as follows:

Corpus Development and Expansion: A standardized Unicode-compliant corpus comprising approximately 100,000 words and 10,196 sentences has been developed by integrating textual data from diverse sources, including literary works, educational materials, folk narratives, cultural publications, Wikipedia, and archival documents. The inclusion of multiple genres improves the representativeness of the corpus.

Unicode Standardization: Approximately 45,000 words from legacy ASCII-based Smriti font documents had been converted into Unicode (UTF-8), enabling interoperability with modern NLP tools and ensuring long-term digital preservation.

Corpus-Based Linguistic Analysis: The developed corpus facilitated quantitative analysis of lexical frequency, morphological variation, and grammatical structures. Approximately 25,000 inflected word

forms have been identified, providing valuable linguistic evidence for future lexicographic and corpus-based studies.

Computational Readiness: The corpus has been processed through cleaning, normalization, sentence segmentation, and tokenization, making it suitable for future NLP applications such as part-of-speech tagging, morphological analysis, language modelling, spell checking, named entity recognition, and machine translation. The study also contributes to the digital preservation of the endangered Bishnupriya Manipuri language by establishing a reusable computational resource for future research.

VII. Conclusion and Future Work

- This paper presents an extended and enriched study on the development of a Bishnupriya Manipuri text corpus, addressing a critical gap in computational resources for the language. By combining traditional linguistic insights with modern corpus development techniques, the study contributes to both language preservation and technological advancement.
- The developed digital language resource establishes a foundation for future NLP research on Bishnupriya Manipuri. Future work will focus on corpus annotation, part-of-speech tagging, morphological analysis, parallel corpus development, and AI-based language technologies such as spell checking and machine translation.

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